

From Physical Space to Service Platform: A Browser-Based Digital Twin Prototype with Live Sensor Overlays and Decision Support

Digital twins are now widely recognised as a key enabling technology for smart services, as they provide a dynamic digital representation of the physical environment that combines spatial models with continuously updated operational data (Grieves & Vickers, 2017). Digital twins use Internet of Things (IoT) data streams and a three-dimensional representation of a physical space to provide real-time monitoring, analysis and decision-making. They are not only used in industrial manufacturing but also in other sectors like smart homes, healthcare, facility management, and education, where they enhance operational efficiency, user experience, and service innovation (Asvadi et al., 2022).

To explore these ideas a prototype is made through the development of a browser-based digital twin of a residential living room. The prototype goes beyond the pure technical implementation and explores the interplay of photorealistic 3D visualisation, contextual sensor information, and decision layers to highlight the potential of digital twins as interactive service platforms. The work fits into an emerging smart service research agenda that investigates the value creation potentials of digital twins for the business and consumer worlds (Nadeem et al., 2025), and the impact of immersive spatial computing platforms on business renewal in service contexts.

From service context, digital twins can minimize the cognitive load and improve situational awareness by integrating data into its spatial context. This spatial approach is quicker than the conventional dashboards for decision making. Digital twins serve as a bridge between the physical world, digital data, and human understanding, bringing them together in a unified interface. This integration has the potential to create novel approaches of value co-creation between users and providers, leading to service innovation, customer engagement, and overall business renewal (Alimamy & Nadeem, 2022; Lyu et al., 2022; Nadeem et al., 2025). This shifts value delivery from selling hardware to delivering outcomes (comfort, energy savings, air quality), consistent with the servitisation paradigm described in manufacturing and IoT literature (Lyu et al., 2022).

For practical demonstration of these concepts, a browser-based digital twin was built using 3D Gaussian Splatting. A multi-stage capturing process was used to reconstruct the physical environment from a living room of a real residence. The final model was created with Scaniverse, manually edited to remove artifacts related to reconstruction work, and rendered in real-time with gaussian-splats-3d library and Three.js. Gaussian Splatting has recently emerged as an extremely efficient alternative to neural rendering techniques like Neural Radiance Fields (NeRF), while preserving high visual fidelity (Kerbl et al., 2023; Mildenhall et al., 2021). Another recent study highlights that Gaussian Splatting is a potential basis for scalable and real-time digital twin systems, due to its computational efficiency and suitability for interactive environments (Chen & Wang, 2024).

To simulate a smart indoor system, five virtual IoT sensors were placed in the 3D reconstructed environment. These include freezer temperature, indoor CO₂ concentration, occupancy, television device state, and temperature. Sensor values are generated using Gaussian noise (Box–Muller transform) combined with simple sequential patterns to approximate realistic environmental variation. Instead of reporting this data in external dashboards, the sensor outputs are visually placed in the 3D scene at their physical locations. By designing this way, users can better understand the environment in real time, further highlighting the spatial characteristic of the digital twin.

A rule-based decision engine continuously checks sensor inputs at regular time intervals and provides recommendations on thermal comfort, air quality, energy efficiency and appliance safety. This is deliberately simple, but it shows an important functional feature of digital twins: turning raw data into actionable service insights. The system does not require continuous human monitoring, rather it highlights smart, situational recommendations for rapid decision making. In the smart home and smart building environment, where the combination of artificial intelligence and the Internet of Things (IoT) is anticipated to enhance predictive maintenance, energy optimisation, and sustainable resource management (Aliero et al., 2021), these decision support functions are becoming more relevant.

The prototype also illustrates the potential of digital twins in the field of sustainable service systems. By integrating continuous environmental monitoring with contextual decision support, digital twins can help optimise energy consumption, improve indoor environmental quality, reduce unnecessary resource usage, and support preventative maintenance strategies. While the current implementation is based on simulated sensor data, the architecture is directly reusable in real-world IoT deployments, enabling evidence-based facility management in residential, educational and commercial applications. This makes digital twins as enablers of sustainability-oriented service innovation. Generally, the prototype represents an iterative service development process that uses low-cost simulations to examine interaction design, decision logic and service value before deploying physical infrastructure. This not only reduces the implementation risk but allows for rapid testing of new service ideas as well. It also shows how spatial computing technologies can be used not just for visualisation, but as core elements for the next generation of service-based ecosystems.

Finally, this prototype redefines digital twins as intelligent service platforms rather than simple visualization tools that combine spatial representation, real-time data, and actionable decision support. The system's photorealistic 3D reconstruction, contextual IoT simulation, and analytics show that the immersive digital environment can significantly contribute to value co-creation, improve business renewal in smart service ecosystems, and support operational decision-making (Nadeem et al., 2025). In the context of the current priorities of smart service research and digital transformation, this work lays the conceptual and technical foundation for further research on digital twins, value creation and business renewal in smart service ecosystems.

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